MPLS LDP TROUBLESHOOTING GUIDE

From Non-Existent to Operational - and why the LSP still does not forward

IOS / IOS-XE / IOS-XR syntax, with Juniper JunOS equivalents where they differ materially

1. NON-EXISTENT — No Hello adjacency → **2. INITIALIZED** — TCP up, Init sent → **3. OPENSENT** — Waiting on peer Init → **4. OPENRECEIVED** — Waiting on KeepAlive → **5. OPERATIONAL** — Labels exchanged

any error -> Hello discovery restarts

NON-EXISTENT

WHY

- LDP not enabled on the interface ("mpls ip" missing)
- ACL blocking UDP 646 Hello or TCP 646 session traffic
- No IGP reachability to the peer transport address

FIX

```
show mpls ldp discovery
show mpls ldp neighbor
verify mpls ip per interface
```

INITIALIZED

WHY

- Firewall blocking TCP port 646
- Wrong transport-address: interface IP vs loopback
- Asymmetric route to the peer transport address

FIX

```
show mpls ldp neighbor detail
telnet <peer-transport-IP> 646
verify transport-address reachability
```

OPENSENT

WHY

- Label space, PVLim, or KeepAlive parameter mismatch
- MD5 authentication mismatch, often silent
- Legacy protocol version mismatch

FIX

```
debug mpls ldp session
verify MD5 passwords exactly
check negotiation errors
```

OPENRECEIVED

WHY

- KeepAlive filtered by a stateful firewall
- Hold timer expires before KeepAlive arrives
- Unstable link resets TCP during negotiation

FIX

```
show mpls ldp neighbor | include Up
compare hold-time configuration
check interface error counters
```

OPERATIONAL

Session up but forwarding broken?

WHY

- FEC has no bound label because the IGP route is missing
- Label filtering blocks advertise or accept behavior
- PHP, explicit-null, or MTU behavior differs from expectation

FIX

```
show mpls forwarding-table
show mpls ldp bindings <prefix>
traceroute mpls ipv4 <prefix>/<mask>
```

VERIFICATION COMMANDS - RUN IN ORDER

1	`show mpls ldp neighbor`	session state per peer
2	`show mpls ldp discovery`	Hello adjacency before session work
3	`show mpls ldp neighbor detail`	hold-time, transport, params
4	`show mpls ldp bindings <prefix>`	local + remote labels for a FEC
5	`show mpls forwarding-table`	LFIB label actions programmed
6	`show mpls interfaces`	MPLS enabled per interface
7	`show ip cef <prefix> detail`	label imposition matches CEF
8	`traceroute mpls ipv4 <prefix>/<mask>`	hop-by-hop LSP validation
9	`ping mpls ipv4 <prefix>/<mask>`	fast LSP liveness check
10	`show mpls ldp parameters`	router-id, transport, hold-time
11	`debug mpls ldp session state-machine`	live negotiation, careful in prod

JUNIPER EQUIVALENTS

`show ldp session | show ldp neighbor | show ldp database | show route table mpls.0 | show rsvp session | traceroute mpls ldp <prefix>`

PHP implicit-null is expected at the penultimate hop, not a missing label.

MPLS LABEL OPERATIONS REFERENCE

PUSH
Ingress LER imposes a transport label; services may add an inner VPN or L2VPN label.

SWAP
Transit LSR replaces the top label with the next-hop label for the same FEC.

POP
Explicit-null pops at egress; implicit-null/PHP pops at the penultimate hop.

LABEL STACK
Outer label crosses the core. Inner service label identifies the VRF or circuit at egress.

TTL MODES
Uniform mode exposes core hops. Pipe mode hides them, so traceroute may appear to jump.

SPECIAL LABELS
0 IPv4 explicit-null | 1 router-alert | 2 IPv6 explicit-null | 3 implicit-null signal

Command clue: show mpls forwarding-table detail | Juniper: show route table mpls.0 extensive

GOLDEN RULES + LDP TIMERS

- LDP needs reachability to the peer transport address before Hello works.
- Enable "mpls ip" per interface; global config alone does nothing.
- Operational session does not prove every FEC has a label.
- Check bindings per prefix before blaming the forwarding plane.
- MD5 auth failures can look exactly like a firewall drop.
- RSVP-TE needs a valid CSPF path and available bandwidth.
- No IGP route to the FEC means no label will be bound.
- Large-packet drops usually point to MPLS MTU along the core.

LDP TIMER REFERENCE

Parameter	Default	Config clue
Hello interval	5s	discovery hello interval
Hello hold-time	15s	discovery hello holdtime
KeepAlive interval	60s	session keepalive behavior
KeepAlive hold-time	180s	mpls ldp holdtime

Session protection can keep bindings briefly across a link flap without tearing down the whole session.

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